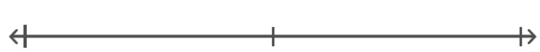
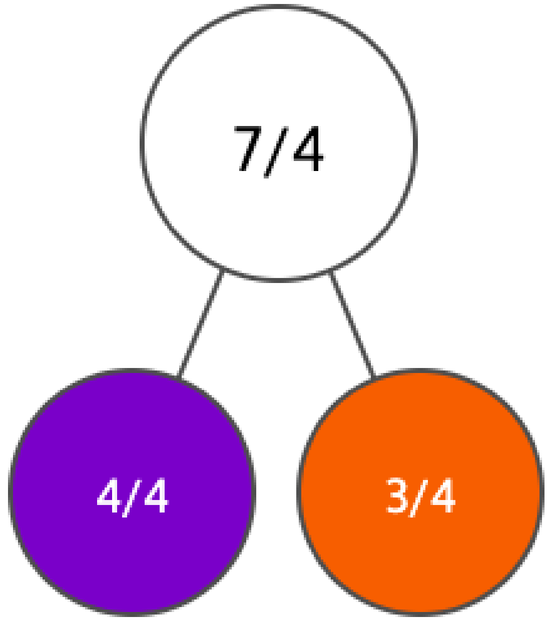
Lesson Objective

Use visual models to add and subtract two fractions with the same units.

Materials: Personal white board

1. Subtract fractions with like units using a number line. Solve 5/6 – 4/6 using a number line.  *Answer: 1/6*	Tell students that just as 5 apples minus 4 apples is 1 apple, 5 sixths minus 4 sixths is 1 sixth. Ask students to say the subtraction sentence using unit form. Guide students through partitioning the number line into sixths and plotting a point at 5/6. Have students count backward by sixths, drawing an arrow above the number line to show subtracting 4/6. Ask where they land. Ask students to write the subtraction sentence in standard form and confirm that 5/6 – 4/6 = 1/6. 
2. Add fractions with like units using a number line. Solve 5/4 + 2/4 using a number line. 	Ask students how they should label the endpoints, since the sum will be greater than 1. Confirm endpoints of 0 and 2, and have students partition the number line into fourths.

Answer: 7/4	Have students plot a point at 5/4 and count forward by fourths to add 2/4, drawing an arrow above the number line. Ask students to say the sum in unit form (5 fourths plus 2 fourths is 7 fourths) and then write the equation in standard form: 5/4 + 2/4 = 7/4. 
3. Use a number bond to rename a fraction greater than 1 as a mixed number. Use a number bond to decompose 7/4 into a whole and a fractional part. Then rename 7/4 as a mixed number. *Answer: 7/4 = 4/4 and 3/4; 7/4 = 1 3/4*	Tell students they can rename a fraction greater than 1 as a mixed number by decomposing it into a whole and a fractional part. Ask students how many fourths are in 1 whole. Confirm that 4/4 = 1. Guide students through drawing a number bond with 7/4 as the whole. Have students record 4/4 as one part and ask how many fourths are left. Have them record 3/4 as the other part. Point out that 4/4 is 1 whole and 3/4 is the remaining fractional part, so 7/4 can be renamed as the mixed number 1 3/4.

Monitor For	Instructional Moves
• Do students partition the number line into equal intervals that match the denominator (sixths in Problem 1, fourths in Problem 2)? • Do students start at the first fraction and count by unit-fractions (1/6, 1/6, 1/6...) rather than by whole numbers when adding or subtracting on the number line? • In Problem 2, do students recognize that the number line needs endpoints of 0 and 2 because the sum is greater than 1? • In Problem 3, do students identify that 4/4 = 1 whole and use that to decompose 7/4 into a whole and a fractional part?	• Reinforce unit-form language by having students restate each problem as a count of unit-fractions (e.g., "5 sixths minus 4 sixths is 1 sixth") before writing the standard-form equation. • If a student counts tick marks instead of intervals when partitioning, prompt them to count the spaces between marks and confirm there are 6 equal parts from 0 to 1 in Problem 1. • In Problem 2, if a student stops the number line at 1, ask how they will plot 5/4 and remind them that 4/4 equals 1 whole, so 5/4 is past 1.

	• In Problem 3, anchor the number bond to the question "How many fourths make 1 whole?" and write 4/4 = 1 next to the bond so students see why one part of 7/4 is 4/4.

Reflection Questions

• How did counting backward by sixths on the number line help you find 5/6 − 4/6? *Answer: Each jump back was 1/6, so after 4 jumps I landed on 1/6.*
• Why did the number line in Problem 2 need to go from 0 to 2 instead of 0 to 1? *Answer: Because 5/4 is already greater than 1, and adding 2/4 makes the sum even larger than 1, so it needs to fit on the line.*
• How did the number bond help you rename 7/4 as a mixed number? *Answer: It showed that 7/4 is made of 4/4, which is 1 whole, and 3/4 left over, so 7/4 is the same as 1 3/4.*
• How is adding and subtracting fractions with like units similar to adding and subtracting whole numbers? *Answer: The unit stays the same, and you just add or subtract how many of that unit you have, like 5 apples minus 4 apples is 1 apple.*